

Engineering Ethics

HUM 4441

Progress Report

Group number : 08

Group Members :

Md. Mahiul Kabir

ID: 220041109

Rahatut Tahrim Mounota

ID: 220041122

Sohom Saboyob Sattyam

ID: 220041141

Imtiaz Morshed Adnan

ID: 220041148

Assignment Title : Ethical responsibilities of engineers and platforms in prioritizing user security, balancing cost and service - A Case Study on Telegram

Timeline

Week 1-2: (COMPLETED)

Identify where trade-offs affect security. Conduct literature review and contextual research (academic papers, tech blogs, news articles, Telegram documentation).

Week 3-4:

Analyze dilemmas using ethical theories and frameworks. Evaluate Telegram's key decisions and balance of stakeholder needs.

Week 5-6:

Finalize written report. Write key sections, develop relevant visuals and seek feedback.

Week 7-8:

Prepare a presentation highlighting key insights, challenges, and recommendations.

Introduction

User security is undeniably one of the most important aspects of any digital platform. The protection of user data, privacy, and access rights fall under its domain.

In this digital age, using apps, websites and other digital platforms requires usage of some level of personal data and so putting trust on these platforms, people share their private data - which if not protected can jeopardize the security and privacy of users. But security often has tangible as well as non tangible costs and can affect how easy or fast a service is made to be used. This creates a challenge of efficiently minimizing the costs without risking the safety of users data.

From an ethical point of view, engineers and platforms must not only protect users and prioritize their good, but also ensure that the service that is being provided does not burden the stakeholders from a cost perspective. Maintaining this intricate balance can lead to taking decisions that may not prioritize the users well being and result in trade-offs.

Telegram is a popular app for messaging that offers privacy, end-to-end encryption and features like large group chats and cloud storage. These makes Telegram suitable for doing a case study on how various digital platforms handle this balance between user security, cost, and service.

This paper looks at the ethical responsibilities of engineers and platforms, using Telegram as a case study.

The Ethical Challenge

In our case study, the ethical challenges we are analyzing are :

- **Privacy vs. Security:** Telegram doesn't use end-to-end encryption by default, it is used for some chats only. This may mislead users about the nature of security of their data. This is an ethical challenge - whether to make privacy a focus or the security.
- **Transparency:** What Telegram does and doesn't do with the data that is being shared by the users, this information must be disclosed from an ethical view - if that is properly maintained or not.
- **Content Moderation:** Malicious content (scams, extremism, pornography) can be shared through the platform, and the actions taken against these can raise questions about its responsibility to act ethically .
- **Security vs. Cost:** Stronger security means higher costs and possibly slower service, which is tough to manage for a free platform.

Engineering Ethics Overview

Engineering Ethics (EE) is the field of study that focuses on the moral principles and professional standards that engineers must follow in their work. It emphasizes a strong commitment to safety, honesty, integrity, and protecting the welfare of people as well as deals with how one must act when facing an ethical dilemma. It is important to maintain proper alignment with the ethical codes of conduct because the decisions taken by professionals in the field impact people, society and public safety.

Organizations like IEEE (Institute of Electrical and Electronics Engineers), NSPE (National Society of Professional Engineers), and ACM (Association for Computing Machinery) provide ethical codes that guide engineers in making responsible choices, especially when facing difficult trade-offs.

Complying with these is crucial in software for safety-critical systems, where a failure can lead to serious harm. In such fields, focusing on ethics isn't optional, it is a must for trust and public safety. In case of prioritizing and maintaining user security engineers must be careful while upholding user rights complying with the legal and ethical standards.

Relevant Materials Chosen For Literature Review :

1. The Security Blanket of the Chat World: An Analytic Evaluation and a User Study of Telegram

(Ruba Abu-Salma^{1,*}, Kat Krol^{2,*}, Simon Parkin¹, Victoria Koh¹, Kevin Kwan¹, Jazib Mahboob¹, Zahra Traboulsi¹, and M. Angela Sasse¹)

2. Investigating User Privacy and Security on Telegram Messenger: A Wireshark Analysis

(Zaid Arafat)

3. Security Analysis of Telegram

(Hayk Saribekyan (hayks@mit.edu), Akaki Margvelashvili (margvela@mit.edu))

4. Automated Symbolic Verification of Telegram's MTProto 2.0

(Marino Miculan, Nicola Vitacolonna)

5. The Tale of Telegram Governance: When the Rule of Thumb Fails

(Farzaneh Badiei)

6. Uncovering the Dark Side of Telegram: Fakes, Clones, Scams, and Conspiracy Movements

(Massimo La Morgia, Alessandro Mei, Alberto Maria Mongardini, Jie Wu)

Observations from the studied papers :

The Security Blanket of the Chat World: An Analytic Evaluation and a User Study of Telegram :

- **Short summary :** This paper studied how real users understand and use Telegram's security features. They ran a lab study with 22 participants (novices and users) and did a usability inspection of Telegram's interface.

Even though people feel safe using Telegram, most of them:

- Use the default chat (which isn't end-to-end encrypted),
- Rarely use Secret Chat (the more secure mode),
- And don't understand how the security features actually work.

The interface doesn't help either , it's full of technical jargon, unclear design, and confuses users about which mode is secure.

- **Identified trade offs:**
 1. People use Telegram thinking it's secure, but most don't realize that regular chats aren't end-to-end encrypted, only Secret Chats are.
 2. The default chats are super convenient and sync across devices, but that ease comes with a cost: your messages are stored on Telegram's servers.
 3. Secret Chats are actually much safer, but they're kind of hidden, a bit clunky to use, and don't work in groups , so most people just skip them.
 4. The app feels clean and simple, but that simplicity can be misleading , it hides how security works, and many users don't know what's actually protected.

Investigating User Privacy and Security on Telegram Messenger: A Wireshark Analysis :

- **Short summary :** This paper takes a close look at how secure Telegram really is, using a tool called Wireshark to track what happens when people send messages, call, share files, or even just stay idle. It is found that Telegram does a decent job with encryption, especially in Secret Chats and voice/video calls, which are truly end-to-end encrypted. However, regular chats (default ones) are only encrypted between the user and Telegram's servers - which means Telegram technically has access. They also noticed that things like contact syncing and profile searches send sensitive data to Telegram servers, and even when you're not actively using the app, it's still pinging the servers in the background. In short, Telegram feels secure and is, in some ways but it depends a lot on how you're using it.

- **Identified trade offs:**

1. Secret Chats and calls are properly encrypted end-to-end, but most people stick to regular chats, which aren't as private as they think.
2. Telegram makes it super easy to sync messages across devices, but that convenience means the messages live on their servers - not ideal if you care about privacy.
3. Contact syncing is helpful when you're setting up the app, but it quietly sends your phonebook to Telegram, which not everyone's cool with.
4. Even when you're not actively using the app, it's still chatting with the servers in the background - which feels a little unnecessary and raises privacy questions.

Security Analysis of Telegram :

- **Short summary :** The paper focuses on a metadata leakage of telegram, termed as "Availability exploit", which addresses the issue of being able to view someone's "online" status without explicit concern.
- **Identified Tradeoff:** Ease of use vs Security enforcements - Often strict security protocols make software less user friendly; while better UX features create vulnerabilities.

Automated Symbolic Verification of Telegram's MTProto 2.0 :

- **Short summary :** The paper conducts a formal verification of MTProto 2.0 which is the cryptographic protocol behind Telegram's secure messaging. Using the Tamarin Prover, a program for proof of cryptographic protocols, the authors analyze secrecy, authentication, and forward secrecy properties of the protocol. They discovered critical flaws in the informal claims made by Telegram and showed that a lot of the security guarantees depend on assumptions that aren't necessarily concrete. Their work presents a formal model of MTProto 2.0 and suggests improvements and guidelines for safer deployment.
- **Identified trade offs :**
 1. Key reuse trade-off : Reusing long-term secret keys or not refreshing keys often leads to loss of forward secrecy, that is when such a key is compromised in the future, past communication still remains secure.
 2. Assumption sensitivity : The protocol's security depends on assumptions that are not enforced by the protocol like clients behaving properly.
 3. Out-of-band verification reliance : Authentication of users often relies on external mechanisms (like phone numbers), which are weak against social engineering or SIM swap attacks.

The Tale of Telegram Governance: When the Rule of Thumb Fails :

- **Short summary :** The paper analyzes the fact that even though Telegram prioritizes user security and privacy as per their motto and the features, there is a lack of clarity in their governance structure, that is, the mechanisms that Telegram, its users and other actors use to govern their interactions with one another. The author has explored the current structure of Telegram and what issues are present in it and through the theory of collective efficacy, provided solutions to the existing problems with the governance structure
- **Identified tradeoffs :**
 1. Security Response Trade-off : There are no details provided on security issue handling and systematic procedure, other than a reporting feature. This minimal disclosure may reduce attack surface awareness but limits user confidence in issue resolution. Delaying fixes exposes users to ongoing threats.
 2. External authority alignment trade-off : Inconsistent attitude towards government alignment has been observed. Resisting overreach may protect autonomy of users in some regions, selective compliance may compromise fairness to users and may put some user groups at risk, especially in surveillance-heavy regions.
 3. Downloadable Channel Content Risk : Allowing channel content to be downloaded can help communities at-risk preserve evidence, but also opens up doors for people with malicious interests to replicate fake channels and spread misinformation more easily.

Uncovering the Dark Side of Telegram: Fakes, Clones, Scams, and Conspiracy Movements:

- **Short summary :** This paper investigates the misuse of Telegram channels by analyzing 35,382 channels and over 132 million messages in a dataset which is called TGDataset. It focuses on three malicious types of channels :
 - a. Scam channels that impersonate public figures or promote fake crypto schemes.
 - b. Clone channels that repost content from popular channels to gain followers and push unrelated links or agendas.
 - c. Fake channels that copy information of official ones but post unrelated or deceptive content.

A machine learning model trained on 342 channels identifies fake ones with 86% accuracy, by using features like volume of link, posting activity, and emojis present in names.

A graph analysis shows scam channels are often just 2–3 hops away from verified ones, which make them easily accessible to the users. Topic modeling reveals that many channels involve illegal or harmful content (e.g., violence, adult content, carding, neo-Nazi propaganda).

- **Identified trade offs:**

1. Telegram's privacy features and open channel creation encourage freedom - but also allow chances of impersonation, fraud, and extremism, risking user security.
2. Scalability comes at the cost of weak moderation and trust signals.
3. Verification systems exist, but users can still easily end up in scam networks through forwarded messages.

Contribution :

1. **Mahiul Kabir (ID 220041109)** : Paper analysis of "Security Analysis of Telegram"
2. **Rahatut Tahrir Mounota (ID 220041122)** : Paper analysis of "The Tale of Telegram Governance: When the Rule of Thumb Fails"
3. **Sohom Saboyob Sattyam (ID 220041141)** : Paper analysis of "The Security Blanket of the Chat World: An Analytic Evaluation and a User Study of Telegram" and "Investigating User Privacy and Security on Telegram Messenger: A Wireshark Analysis"
4. **Imtiaz Morshed Adnan (ID 220041148)** : Paper analysis of "Automated Symbolic Verification of Telegram's MTPROTO 2.0" and "Uncovering the Dark Side of Telegram: Fakes, Clones, Scams, and Conspiracy Movements"